

Atemporality of Mental Memory Space

A Structural Hypothesis Grounded in Resource Constraints,
Cyclical Closure, and Reconstructive Retrieval

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Abstract

This paper develops a structural hypothesis about memory architecture: that temporal ordering is not an intrinsic property of memory state spaces but emerges through retrieval operations. Unlike standard reconstructive theories that treat temporal flexibility as a psychological phenomenon to be explained, this account proposes atemporality as a *structural consequence* of resource constraints and cyclical closure conditions.

We show how memory state spaces *can be modeled* as cyclically closed structures satisfying fixed-point conditions, and argue that for systems meeting these conditions, the question “which memory came first?” may represent a conceptual mismatch—analogueous to asking “what is north of the North Pole?” The framework draws on the Evolutionary Theory of Credence (resource constraints necessitating commitment beyond verification) and formal results on self-sufficient structures (fixed-point existence, topological closure).

The central thesis is **modal and conditional**: *if* memory systems satisfy specified closure and resource conditions, *then* their state space architecture is atemporal in the sense that temporal ordering attaches to operations rather than states. This framework offers higher explanatory coherence for known phenomena (temporal distortions, reconsolidation effects, cross-temporal associations) than linear-indexing alternatives, while generating testable predictions distinguishing it from weaker reconstructive accounts.

Scope limitation: This paper establishes formal conditions under which atemporal architecture is coherent and advantageous, not that biological memory necessarily instantiates this specific structure. The contribution is a structural explanatory model with empirical implications, not a proof of memory architecture.

Keywords: atemporality, memory architecture, cyclical hierarchy, fixed-point theory, resource constraints, reconstructive retrieval, structural hypothesis

1 Introduction

1.1 The Problem of Temporal Priority in Memory

Traditional models of memory implicitly assume a linear, temporally ordered archive: memories are created at specific times, stored with temporal indices, and retrieved by accessing time-stamped records. This conception faces both empirical and theoretical difficulties.

Empirically: Memory exhibits remarkable temporal flexibility—childhood memories can be as vivid as yesterday’s events; associations form across vast temporal distances; retrieval systematically modifies what is retrieved [Nader et al., 2000]; temporal distortions (telescoping, boundary effects, displacement) are pervasive rather than exceptional [Friedman, 1993].

Theoretically: Strict temporal indexing would impose storage and computational costs that scale poorly with system complexity—a concern formalized by resource-theoretic approaches to bounded cognition [Simon, 1957, Gigerenzer and Selten, 2001].

1.2 The Structural Hypothesis

This paper develops an alternative: memory storage *can constitute* a cyclically closed structure in which temporal ordering is not intrinsic to the state space but emerges through retrieval operations.

Hypothesis 1 (Atemporality of Memory State Space). *For memory systems satisfying:*

1. *Resource constraints limiting verification capacity relative to state space size*
2. *Cyclical closure conditions (associative relations forming closed paths)*
3. *Reconstructive retrieval (access modifies and rebuilds content)*

*temporal ordering attaches to **operations** (encoding, retrieval) rather than **states** (stored representations). Temporal experience is constructed during retrieval, not read from storage.*

Critical distinction from standard reconstructive theories: Accounts following Bartlett [1932] and Schacter et al. [2012] treat temporal distortion as a *psychological fact* requiring explanation. The present account treats atemporality as a *structural consequence* of architecture under specified constraints—explaining *why* memory must be reconstructive rather than merely *that* it is.

1.3 What This Paper Does and Does Not Claim

Claims:

- We *can model* memory state spaces as cyclically closed structures (formal modeling claim)
- For systems satisfying specified conditions, temporal ordering emerges from operations rather than states (conditional structural claim)
- This architecture has higher explanatory coherence for known phenomena than linear-indexing alternatives (comparative explanatory claim)
- The framework generates distinguishing predictions (empirical fertility claim)

Does not claim:

- That biological memory *necessarily* instantiates this specific structure
- That the topological formalization directly describes neural implementation
- That temporal information is entirely absent from memory (it exists as *content*, not *structure*)
- That this framework uniquely explains the phenomena (alternatives may also explain them)

1.4 Relation to Existing Frameworks

Reconstructive memory theories [Bartlett, 1932, Loftus, 1979, Schacter et al., 2012]: These establish that memory is constructive and error-prone. The present framework explains *why* this is so architecturally, rather than treating it as contingent psychological fact.

Temporal Context Model [Howard and Kahana, 2002]: TCM encodes temporal *relations* (proximity via context similarity) rather than temporal *positions* (absolute timestamps). This is consistent with our framework: temporal information as distributed content feature, not

structural axis. However, TCM does not address whether this reflects fundamental architectural constraints or implementation choice.

Predictive processing [Clark, 2013, Friston, 2010]: These accounts emphasize prediction and model-based processing. The present framework adds that memory storage itself may be atemporal, with temporal structure emerging through predictive reconstruction.

Grounding and fundamentality [Schaffer, 2009, Barnes, 2018, Bliss and Priest, 2018]: Debates about symmetric or non-well-founded grounding provide philosophical precedent for cyclical dependency structures. We do not propose revising grounding theory but note structural parallels.

Episodic vs. semantic memory [Tulving, 1972]: The atemporality hypothesis applies most directly to the storage architecture underlying both systems, not to the phenomenology of episodic recall. Episodic memory’s “what-where-when” character reflects content and retrieval processes, not necessarily storage structure.

1.5 Paper Organization

Section 2 develops the resource-constraint argument. Section 3 presents the formal framework. Section 4 discusses retrieval-based temporal construction. Section 5 examines empirical alignment and predictions. Section 6 explores implications. Section 7 addresses objections.

2 Resource Constraints and Verification Limits

2.1 The Verification-Complexity Trade-off

Complex adaptive systems face a fundamental constraint: as system complexity increases, the space of action-relevant distinctions expands faster than verification capacity can grow [Simon, 1957, Gigerenzer and Selten, 2001].

Definition 1 (Verification Coverage—Scaling Relation). *For an adaptive system S , verification coverage scales approximately as:*

$$V(S) \sim \frac{B(S) \cdot \tau}{|\mathcal{P}(S)|} \quad (1)$$

where $B(S)$ is verification bandwidth (rate of direct confirmation), τ is decision latency, and $|\mathcal{P}(S)|$ is a proxy for decision-relevant distinction space size.

Clarification: Here “ $\sim$ ” signals a scaling relation, not exact equality. The point is that verification capacity grows at most linearly in $B(S)$ and τ , while $|\mathcal{P}(S)|$ grows combinatorially with complexity (planning depth, social reasoning, behavioral repertoire).

As complexity increases, $V(S) \rightarrow 0$: complex systems cannot verify most action-relevant propositions at decision time. This is the foundation of the Evolutionary Theory of Credence [Kriger, 2024]: complex systems must commit to propositions exceeding immediate evidential warrant.

2.2 Memory as Credal Commitment

Memory retrieval exemplifies commitment beyond verification:

Principle 1 (Memory as Credence). *Every act of **internal biological memory** retrieval involves commitment to propositions that cannot be verified at retrieval time without external records:*

- *The retrieved content accurately represents past states*
- *The retrieval process has not distorted the content*

- *The temporal ordering attributed to memories reflects actual sequence*

This is not a deficiency but a structural consequence: a system requiring verification of all memory content before use could not function on adaptive timescales.

2.3 The Cost of Temporal Indexing

Proposition 1 (Resource Argument Against Comprehensive Temporal Indexing). *Comprehensive temporal indexing of all memory states faces scaling difficulties for complex systems.*

Argument (qualified):

1. Temporal indexing requires storing temporal metadata with each memory state
2. For systems with many states, maintaining temporal structure requires either:
 - Absolute timestamps (requiring synchronization infrastructure and storage)
 - Relative ordering (whose maintenance cost depends on implementation)
3. Under bounded resources, comprehensive temporal indexing becomes increasingly costly as state space grows

Important qualification: This argument establishes that comprehensive temporal indexing is *costly*, not *impossible*. Temporal structure can be implicit and distributed (as in TCM’s context drift) rather than explicit metadata. The claim is not that temporal information is absent, but that storing it as a *structural axis* of the state space (like coordinates in a temporal database) faces resource pressures that storing it as *content features* does not.

The key distinction: Temporal information encoded as content (“this happened before that”) differs from temporal structure as architecture (memory state s occupies position t in a temporally-ordered space). The former is compatible with atemporal state spaces; the latter is not.

3 Formal Framework

3.1 Preliminary: The Foundational Mismatch Concept

Before presenting formal definitions, we introduce the conceptual target:

Definition 2 (Conceptual Mismatch). *A conceptual mismatch occurs when concepts appropriate to one type of structure are applied to a structure lacking the features those concepts presuppose.*

Paradigmatic example: “What is north of the North Pole?” The question is grammatically well-formed but geographically inapplicable—it applies “north of” beyond its domain. Similarly, we will argue that “which memory is temporally first in the state space?” may apply temporal ordering beyond its domain for certain memory architectures.

3.2 Memory State Spaces

Definition 3 (Memory State Space). *A memory state space is a triple $(\mathcal{S}, \mathcal{R}, \mathcal{F})$ where:*

- $\mathcal{S}$ is a non-empty set of memory states (stored representations)
- $\mathcal{R} \subseteq \mathcal{S} \times \mathcal{S}$ is an association relation
- $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ is a retrieval-reconstruction function

Interpretive note: This is a *modeling framework*, not a claim that memory literally consists of set-theoretic objects. The framework captures structural features (states, associations, retrieval dynamics) abstractly.

Definition 4 (Cyclical Closure). *A memory state space is cyclically closed of order n if there exist states $s_1, \dots, s_n \in \mathcal{S}$ and functions $F_1, \dots, F_n$ such that:*

1. $s_i = F_i(s_{i+1})$ for $i = 1, \dots, n-1$
2. $s_n = F_n(s_1)$
3. $(s_i, s_{i+1}) \in \mathcal{R}$ for all i (indices mod n)

Intuition: Cyclical closure captures mutual determination—memory A is associated with B , B with C , C with A , forming a closed loop with no distinguished “first” element.

3.3 Fixed-Point Existence

Proposition 2 (Fixed-Point Equivalence). *A cyclically closed memory space of order n exists if and only if the composite operator $\Phi = F_1 \circ F_2 \circ \dots \circ F_n$ has a fixed point.*

Proof. By substitution: $s_1 = F_1(F_2(\dots F_n(s_1)\dots)) = \Phi(s_1)$. Conversely, if $\Phi(s_1) = s_1$, the cycle can be recovered. $\square$

Intuition: Walking once around the associative cycle returns you to the same state—this is the fixed point of the composite operator.

The following classical theorems provide sufficient conditions for fixed-point existence:

Theorem 1 (Knaster-Tarski). *If $(L, \leq)$ is a complete lattice and $\Phi : L \rightarrow L$ is order-preserving, then Φ has fixed points.*

Theorem 2 (Banach Contraction). *If (X, d) is a complete metric space and $\Phi : X \rightarrow X$ is a contraction, then Φ has a unique fixed point.*

Remark 1 (Scope of Fixed-Point Results). *These theorems establish existence under general conditions. They apply to many systems beyond memory—which is a feature, not a bug. The claim is not that memory is uniquely characterized by fixed-point structure, but that memory can belong to this class of systems, with the architectural consequences that entails.*

3.4 Topological Characterization

We can characterize when cyclical structure is intrinsic rather than accidental using algebraic topology.

Definition 5 (Essential Cyclicity). *Let X be a topological space encoding memory structure. X is essentially cyclic if:*

1. X is compact and connected
2. The first homology group $H_1(X; \mathbb{Z}) \neq 0$

Proposition 3. *If X is essentially cyclic, the cyclic structure cannot be continuously deformed away—there is no retraction to a point preserving structure.*

Remark 2 (Encoding Dependence—Critical Limitation). *The homological characterization depends on how memory structure is encoded as topological space. Different encodings may yield different homology groups. This is acknowledged as a limitation.*

- The topological framework currently functions as structural metaphor providing precise language for “irreducible cyclicity”
- To become directly applicable formalism, concrete encoding schemes for memory systems must be specified
- The key claim is **conditional**: if a memory system admits encoding with non-trivial H_1 , then its cyclicity is essential and temporal-primacy questions are structurally inapplicable

3.5 Illustrative Encoding: Associative Graph

To make the topological framework more concrete, consider a simplified encoding:

Definition 6 (Associative Graph Encoding). *Given memory states $\mathcal{S}$ and associations $\mathcal{R}$, construct simplicial complex X with:*

- 0-simplices (vertices): elements of $\mathcal{S}$
- 1-simplices (edges): pairs $(s_i, s_j) \in \mathcal{R}$

For this encoding, $H_1(X; \mathbb{Z}) \neq 0$ iff the association graph contains cycles not contractible to points—i.e., closed associative loops.

Example: Three memories $\{A, B, C\}$ with associations $A \leftrightarrow B$, $B \leftrightarrow C$, $C \leftrightarrow A$ form a triangle. This has $H_1 \cong \mathbb{Z}$ —one independent cycle. In this structure, asking “which memory is first” has no distinguished answer; each is equally positioned.

3.6 The States-Operations Distinction

Definition 7 (Memory State). *A memory state $s \in \mathcal{S}$ is a stored representation. States possess identity independent of temporal parameters: state s is identifiable as s regardless of when encoded or retrieved.*

Definition 8 (Memory Operation). *A memory operation is a process acting on states:*

- **Encoding**: $enc_t : \mathcal{E} \times \mathcal{S} \rightarrow \mathcal{S}$ (experience + current state $\rightarrow$ updated state)
- **Retrieval**: $ret_t : \mathcal{Q} \times \mathcal{S} \rightarrow \mathcal{S} \times \mathcal{O}$ (query + state $\rightarrow$ modified state + output)

Operations are indexed by time t . Temporality attaches to operations, not states.

Principle 2 (Atemporality Principle—Conditional Formulation). *In memory systems where stored states are defined independently of creation/access events, and associations form cyclically closed structures, temporality applies to memory **operations** and not to the memory state space itself.*

Contrast with strong temporal-encoding view: On strong temporal-encoding views, time is part of the coordinate system of memory space—like a temporal database where each record has a time-coordinate. On the present view, temporal coordinates, where present, are attached to operations or encoded as content features, not built into state-space architecture.

4 Mechanism: Retrieval-Based Temporal Construction

4.1 Reconstructive Retrieval

Research on reconsolidation [Nader et al., 2000, Sara, 2000] demonstrates that retrieval is not passive reading but active reconstruction:

Principle 3 (Reconstructive Retrieval). *Each retrieval operation:*

1. *Renders accessed memory labile (susceptible to modification)*
2. *Reconstructs content from stored patterns and current context*
3. *Re-encodes the (potentially modified) result*
4. *Updates associative relations based on retrieval context*

This implements the credal commitment that resource constraints necessitate: the system commits to reconstructed content without verifying accuracy against original encoding.

4.2 Temporal Construction Process

If temporal ordering is not intrinsic to states, how does temporal experience arise?

Definition 9 (Temporal Construction). *Temporal ordering in memory emerges from:*

- **Temporal metadata:** *Where explicitly encoded, temporal information functions as content (“summer 2020”), not structure*
- **Contextual reconstruction:** *Temporal context inferred from associated content features (clothing, technology, companions)*
- **Causal inference:** *Temporal order deduced from causal relationships (“I graduated before starting that job”)*
- **Familiarity gradients:** *Subjective “oldness” based on accessibility and vividness*
- **Source monitoring:** *Attribution of “when” based on phenomenal qualities [Johnson et al., 1993]*

Key prediction: Temporal ordering should be more accurate when supported by rich contextual and causal information, and more prone to error when relying on phenomenal qualities alone. This aligns with findings on temporal memory [Friedman, 1993].

4.3 Memory Attractors

Stable memory configurations are fixed points of retrieval-reconstruction:

Definition 10 (Memory Attractor). *A memory configuration $s^* \in \mathcal{S}$ is an attractor if $\mathcal{R}(s^*) = s^*$ —retrieval does not change it.*

This explains why frequently retrieved memories become stable (convergence to fixed points) while enabling “memory modifications”—changes when the system settles into different attractors.

5 Empirical Alignment and Predictions

5.1 Explanatory Coverage

The framework provides unified explanations for diverse phenomena:

Phenomenon	Explanation under Atemporality Hypothesis
Temporal distortions (telescoping, displacement)	Temporal order is reconstructed, not read; reconstruction uses heuristics that can err
Cross-temporal associations	No temporal barriers in storage; associations based on content similarity, not temporal proximity
Reconsolidation effects	Retrieval reconstructs and re-encodes; modification is intrinsic to the process
Childhood amnesia	Early memories contextually disconnected from current retrieval patterns, not temporally distant in storage
Flashbulb memories	Strong contextual encoding provides rich reconstruction cues; vividness reflects context strength, not different storage
Source monitoring errors	Temporal attribution based on inference, not temporal address lookup

Table 1: Explanatory coverage of the atemporality hypothesis

5.2 Comparison with Alternatives

Vs. Linear indexing models: Linear models predict temporal errors should reflect corruption/degradation of temporal indices. The atemporality model predicts errors should reflect inference failures in reconstruction—different error patterns for the same phenomena.

Vs. Weaker reconstructive accounts: Standard reconstructive accounts (Bartlett, Schacter) establish *that* memory is constructive. They are compatible with either:

1. Atemporal storage + reconstructive retrieval (our account)
2. Temporal storage + imperfect/reconstructive retrieval

Distinguishing prediction: Under (1), temporal structure should be systematically reconstructed from non-temporal cues (context, causation). Under (2), temporal structure should be partially preserved with degradation. Empirical tests could examine whether temporal accuracy correlates with temporal-index preservation (favoring 2) or with contextual/causal richness (favoring 1).

5.3 Alignment with TCM

The Temporal Context Model [Howard and Kahana, 2002] encodes temporal relations through gradually drifting context, not explicit timestamps. Items encoded close in time share similar contexts.

Interpretation: TCM encodes temporal *relations* as content (context similarity), not temporal *positions* as structure. This is consistent with—and arguably an implementation of—the atemporality hypothesis. The context drift mechanism distributes temporal information across content features rather than maintaining a temporal axis in state space.

5.4 Time Cells and Specialized Mechanisms

Hippocampal time cells [MacDonald et al., 2011] fire at specific temporal intervals during delay periods, encoding temporal structure.

Interpretation: The existence of specialized temporal-encoding mechanisms *supports* rather than contradicts the hypothesis. If temporal ordering were automatic/intrinsic to memory architecture, dedicated mechanisms would be unnecessary. Time cells represent a specialized system for tasks requiring precise temporal encoding, operating *on top of* a default architecture that lacks intrinsic temporal structure.

Analogy: Color vision requires specialized photoreceptor systems; this doesn’t mean visual cortex is “intrinsically colored.” Similarly, temporal encoding requires specialized mechanisms; this doesn’t mean memory state space is “intrinsically temporal.”

5.5 Testable Predictions

1. **Contextual dependence of temporal accuracy:** Temporal ordering accuracy should correlate with richness of contextual/causal information, not with temporal distance per se.
2. **Interference patterns:** Memories with similar contexts but different times should show more temporal confusion than memories with different contexts but similar times (context dominates over temporal proximity in retrieval).
3. **Reconstruction signatures:** Temporal ordering judgments should show signatures of inference (response time patterns, confidence calibration) rather than direct access.
4. **Developmental trajectory:** Temporal accuracy should improve with development of contextual encoding and causal reasoning, not simply with maturation of a “temporal encoding system.”

6 Implications

6.1 For Artificial Intelligence

Modern AI architectures exhibit parallel structure:

- **Transformer models:** Store representations in embedding spaces without intrinsic temporal ordering; positional encodings add temporal information as content features
- **Vector databases:** Knowledge stores where temporal metadata is optional attribute, not structural axis
- **Retrieval-augmented generation:** Context reconstructed at query time from atemporal storage

Principle 4 (AI Memory Design Implication). *The framework suggests:*

1. *Default to atemporal storage unless task analysis demonstrates necessity for temporal structure*
2. *Encode temporal information as content features rather than structural dimensions*
3. *Implement reconstruction at retrieval rather than maintaining temporal indices*
4. *Add specialized temporal mechanisms only where tasks require precise temporal encoding*

Reframing “hallucination”: AI confabulation arises from the same generative reconstruction that enables useful generalization. Under atemporal storage with reconstructive retrieval, some outputs will have incorrect temporal ordering or content—not through malfunction but through the architecture that enables flexibility. The goal is calibration (knowing when reconstruction is reliable), not elimination (which would eliminate useful capability).

6.2 For Cognitive Science

The framework unifies phenomena often treated separately:

- Temporal distortions, reconsolidation, cross-temporal association, and source monitoring all reflect the same underlying architecture
- Specialized temporal encoding (time cells, sequence learning) represents task-specific augmentation of a default atemporal system

6.3 Speculative Extensions

The following implications are speculative—marked as such and not central to the argument:

Personal identity: If memory storage is atemporal, narrative self-continuity is an achievement of retrieval processes, not a feature of storage. The “self across time” is constructed, not stored.

Temporal experience: The phenomenal “flow of time” in remembering may be a constructed feature of retrieval, not a read-out of temporal structure in storage.

These extensions go beyond what the formal framework directly supports and should be evaluated independently.

7 Objections and Replies

Objection 1: Episodic memory is inherently temporal.

Reply: Episodic memory includes temporal information, but this can be encoded as *content* (“this happened in summer”) rather than *structure* (memory occupies position t in temporal space). The phenomenology of episodic recall—“re-experiencing” an event with temporal context—is consistent with temporal information being reconstructed during retrieval from content features. The atemporality claim concerns storage architecture, not the phenomenology of recall.

Objection 2: Time cells prove temporal structure in memory.

Reply: Time cells demonstrate that specialized mechanisms exist for temporal encoding in specific tasks (delay periods, sequence learning). This is consistent with—and arguably supports—the view that the default architecture lacks intrinsic temporal structure, requiring specialized systems for tasks where temporal precision matters. See Section 5.4.

Objection 3: This is merely mathematical possibility, not psychological reality.

Reply: Correct—and explicitly acknowledged. The paper establishes: (a) formal conditions under which atemporal architecture is coherent; (b) resource-theoretic arguments for why such architecture is advantageous; (c) explanatory alignment with known phenomena; (d) distinguishing predictions. Empirical adequacy is to be determined by investigation, not claimed a priori. The contribution is a *structural explanatory model*, not a proof of architecture.

Objection 4: The topological framework is unfalsifiable given encoding freedom.

Reply: Acknowledged as a current limitation (Remark 4). The topological apparatus provides precise language for structural concepts but requires concrete encoding schemes to become directly testable. Section 3.5 provides an illustrative encoding; developing empirically grounded encodings is future work. The core claims do not depend on the topological formalization—it strengthens but is not necessary for the argument.

Objection 5: Temporal structure can be implicit and distributed, not just meta-data.

Reply: Correct and important. TCM shows temporal relations can be encoded through context similarity without explicit timestamps. This is *consistent with* the atemporality hypothesis: temporal information encoded as distributed content features (context similarity) differs from

temporal structure as a coordinate axis of state space. The claim is not that temporal information is absent but that it functions as content, not architecture.

Objection 6: The photograph analogy is misleading—encoding is dynamic.

Reply: Fair point. Photographs are static artifacts; memory encoding involves dynamic, recursive processes. The analogy illustrates only that causal origin in time does not entail temporal structure in the product. A better analogy: a melody has temporal structure in performance but can be stored as atemporal notation (pattern of intervals). The notation has no “first note in storage”—the sequence emerges in performance. Similarly, memories may be stored as patterns whose temporal structure emerges in retrieval-performance.

8 Conclusion

This paper has developed a structural hypothesis about memory architecture: for systems satisfying resource constraints, cyclical closure conditions, and reconstructive retrieval, temporal ordering is not intrinsic to the state space but emerges through retrieval operations.

Central results:

1. **Resource grounding:** Complex systems face verification-complexity trade-offs making comprehensive temporal indexing costly; temporal information as content is cheaper than temporal structure as architecture
2. **Formal characterization:** Memory state spaces can be modeled as cyclically closed structures satisfying fixed-point conditions; for essentially cyclic spaces, temporal-primacy questions may be conceptual mismatches
3. **Mechanistic implementation:** Reconstructive retrieval implements credal commitment while constructing temporal ordering from content features
4. **Empirical alignment:** The framework provides unified explanations for temporal distortions, reconsolidation, cross-temporal association, and source monitoring, while generating distinguishing predictions

Status of the contribution: This is a *structural explanatory model*—a formal framework that, if memory systems satisfy the specified conditions, explains why temporal flexibility is architecturally natural rather than anomalous. Whether biological memory instantiates this specific structure is an empirical question the framework helps to investigate.

The doctrine “atemporality of mental memory space” emerges not as proof but as hypothesis: if resource constraints, cyclical closure, and reconstructive retrieval characterize memory systems, then temporal ordering is a constructed achievement rather than a stored given. Time belongs to the *use* of memory, not to memory itself.

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